Yashar Zafari Haqqi

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Education

Sharif University of Technology **Tehran, Iran**

M.Sc. in Mechatronics Engineering Sep. 2025 – Present (expected Jul. 2027)

Thesis: "*Learning to Interpret and Execute Iranian Sign Language via Skill-Based Deep Reinforcement Learning: An Offline-to-Online, Cross-Lingual Framework for Robotic Deployment.*"
Advisors: Prof. Alireza Taheri and Prof. Ali Meghdari

Sharif University of Technology **Tehran, Iran**

B.Sc. in Mechanical Engineering, GPA: 18.13/20 (3.84/4) Sep. 2020 – Jul. 2025

Dynamic Systems, Control & Robotics specialization

Thesis: "*Simulation of Dexterous Robotic Hand Object Grasping Control Using Reinforcement Learning.*"
Advisor: Prof. Alireza Taheri

Research Interests

- Physical AI, Robotics, Deep Reinforcement Learning, Dexterous Manipulation, Learning-Based Control

Academic Experience

Research Assistant **Dec. 2023 – Present**

Social & Cognitive Robotics Lab, Sharif University of Technology

- **M.Sc. thesis:** built the simulation of the 36-actuator RASA humanoid and a library that retargets human sign-language motion (3.5M frames, 6 datasets) onto it. On held-out signers, a sign classifier recognizes RASA's replays at **75% of its accuracy on human video**, the ceiling for the RL executor now in development.
- **B.Sc. thesis:** built a grasping pipeline for a 26-DoF Shadow Hand on a UR10e in MuJoCo, combining GraspAnything grasp-point detection, trajectory planning, and PPO finger control: **88% success on training shapes, 69% on 8 unseen objects**. Code and models open-source.

Teaching Assistant **Feb. 2022 – Present**

Department of Mechanical Engineering, Sharif University of Technology

- TA for **Robotics**, **Robotics Lab**, **Aerial Robotics**, **Dynamics of Machinery**, **Dynamics**, **Statics**, and **Measurement & Control Systems Lab** across 10 semesters, under Prof. Alireza Taheri, Prof. Saeed Behzadipour, and Dr. Amin Talaeizadeh.

Honors

National Graduate University Entrance Exam **Feb. 2025**

Ranked **1st** in Mechatronics Engineering group.

National Olympiad - Mechanical Engineering **Oct. 2024**

Ranked **5th** and earned a fellowship to be admitted to any national university without entrance exam.

National Undergraduate University Entrance Exam **Summer 2020**

- Ranked **333rd** among more than 155,000 participants in Math-Physics group.
- Ranked **97th** among more than 174,000 participants in Foreign Languages group.

Papers

Y. Zafari, MJ. Shamseddin, Y. Tahmasebzadeh, "R2C2R: Risk-Sensitive Riccati-Structured Critic with Certified Robustness." *arXiv preprint in preparation*, 2026.

Related Coursework

- **M.Sc.:** Reinforcement Learning, Advanced Topics in Machine Learning: Multi-Agent RL (in progress), Robust Control (in progress), Convex Optimization.
- **B.Sc.:** Robotics, Aerial Robotics, Signals & Systems, Automatic Control, Vibrations.

Projects

R2C2R: Risk-Sensitive Riccati-Structured Critic with Certified Robustness **Fall 2026**

Lead author | Python, NumPy/SciPy, PyTorch | Preprint in Preparation

- Learns risk-sensitive control as a principal eigenpair with a Riccati-structured RL critic, then certifies two-sided bounds on the regional risk-sensitive rate that remain valid beyond the breakdown point, computed with outward rounding. Validated on the inverted pendulum and cart-pole.

RoboPose: Multi-Robot Pose Estimation from Synthetic Data **Fall 2026**

Individual | PyTorch, Ultralytics YOLO, MuJoCo | In Progress

- One real-time model that detects robots of four classes (humanoid, quadruped, arm, quadcopter; 34 models) and their landmarks, trained only on randomized MuJoCo renders. Synthetic validation: pose mAP 0.61, precision 0.99, recall 0.94; real-image evaluation in progress.

ROS2 Driver for RASA Robot's New Hand **Spring 2026**

Individual | ROS2 (C++/Python), MuJoCo | Paused: hardware under repair

- ROS2 package to drive RASA's new hand, as the basis for transferring the B.Sc. grasping policy from simulation to hardware.

Employment

CEDRA Robotics **Nov. 2025 – Present**

Robotics Software Engineer | *Part-time* Tehran, Iran

- Developing the software stack for Taban, a social human-interactive on-table robot.

Ramooz Group **Mar. 2025 – Sep. 2026**

Technical Project Lead, Robotic Inspection | *Part-time* Tehran, Iran

- Led the design of a semi-automated robotic inspection system (underwater ROV with a companion uncrewed surface vessel) for offshore fish farms in high-current waters.

Internship

Iran Tractor Manufacturing Company (ITMCO) **Jun. 2023 – Sep. 2023**

Summer Internship Tabriz, Iran

- Diagnosed and resolved CNC machine faults in the Electronic Maintenance & Repair group, working with industrial servomotors, controllers, sensors, and PLCs.

Technical Skills

Programming: Python (NumPy, SciPy, PyTorch, TensorFlow), C++, C, MATLAB | **Tools**: Git, Linux, LaTeX
Robotics & Simulation: ROS2, MuJoCo, Gymnasium, Genesis, Gazebo | **CAD**: SolidWorks, AutoCAD, CATIA
Learning: Stable Baselines3, Ultralytics YOLO | **Embedded**: Arduino, Raspberry Pi

English Proficiency

TOEFL iBT **Oct. 2024**

Overall: **112/120**, Reading: 29, Listening: 30, Speaking: 24, Writing: 29.

References

Prof. Alireza Taheri
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